Palaeomagnetism, Potassium–Argon Ages and Petrology of some Volcanic Rocks

by
G. BRENT DALRYMPLE
ALLAN COX*
US Geological Survey,
Menlo Park,
California

Geological evidence obtained from samples of rocks from Cocos Island, Costa Rica, seems to agree with the idea that this island could have been a stepping stone for organisms migrating to the Galapagos Islands.

Cocos Island lies in the eastern Pacific Ocean on the crest of the Cocos Ridge, a well defined linear bathymetric high with an average depth at its crest of about 1,000 fathoms[1]. Extending south-west from Costa Rica to the Galapagos Islands, where it joins the Galapagos Fracture Zone, Cocos Ridge appears to be a fracture zone similar to the Tehuantepec Ridge, which branches from the Clipperton Fracture Zone, and to the Nasca Ridge, which branches from the Easter Island Fracture Zone. The age of Cocos Island is of geophysical interest as an indication of the age of the volcanism associated with the fracture zone, and as a measure of the minimum age of the sea floor in this tectonically active region. The age is also of biogeographical interest because of the possibility that Cocos Island served as a stepping stone for plants and animals migrating to the Galapagos Islands from Central America. The visit of the Galapagos International Scientific Project to Cocos Island on March 8–9, 1964, gave one of us (A. C.) an opportunity to collect rock samples in the vicinity of Chatham Bay (Fig. 1). The results of potassium–argon age determinations, palaeomagnetic measurements and chemical analyses are reported here.

In spite of more than 400 treasure-seeking and scientific expeditions to Cocos, surprisingly little is known about its geology. Merrill[2] published a description and partial chemical analysis of an "andesite" collected by Agassis from the vicinity of Chatham Bay. Chesterman[3] described two olivine basalts, one latite, one lapilli tuff-breccia and one "oligophyre" composed mostly of oligoclase, collected on earlier expeditions. The most complete report of the geology of Cocos is that of Chubb[4], who spent 24 h there in 1924. He noted that the island was composed chiefly of massive lava flows and described an agglomeratic tuff from the south side of Wafer Bay. From a study of beach pebbles collected near Wafer Bay, he concluded that the rocks there are principally labradorite–andesite and hornblende labradorite–andesite. Chubb presented chemical analyses of two typical specimens and described them as similar to those of the Galapagos and especially of Rapa. Macdonald[5], however, recognizing the inadequacy of the geological information from Cocos Island, did not include the Cocos rocks in the mid-Pacific volcanic province.

The botanist Alban Stewart found that the flora of Cocos was that of a typical oceanic island, and that there was no botanical evidence for a land bridge connecting Cocos with the Galapagos Islands[6]. He concluded that Cocos Island was younger than the Galapagos Islands because the percentage of endemic plants is smaller, an argument that seems tenuous in view of the fact that Cocos is only half as far from the mainland, has a different climate and is in different oceanic and atmospheric current systems. Vinton inferred a Pleistocene age for the island from the thinness of the soils, the ungraded character of the streams and the scarcity of beach deposits[7]. He further concluded that, during the Oligocene and Miocene, Cocos Ridge was a peninsula extending from Central America to within less than 200 km of the Galapagos Islands but that the peninsula submerged well before the formation of Cocos Island. Channels have been discovered on the sides of Cocos Ridge, but Wilde[8] concluded that these were formed by turbidity currents rather than by subaerial erosion. Furthermore, the crust underlying the ridge seems to be oceanic rather than continental[9], and the absence of guyots on the ridge indicates that it has remained submerged at least since the time of the volcanic

Table 1. CHEMICAL ANALYSES AND MOLECULAR NORMS OF TRACHYTE (4C400) AND ALKALI BASALT (4C401, 4C402 AND 4C404) FROM COCOS ISLAND

	4C400	4C401	4C402	4C404	Average oceanic[12] alkali basalt
SiO_2	58·4	48·3	48·8	48·6	47·4
Al_2O_3	19·0	19·0	18·6	19·1	18·0
Fe_2O_3	4·7	6·5	5·3	8·3	4·2
FeO	1·9	2·8	3·7	1·1	5·8
MgO	0·69	3·7	3·8	2·6	4·8
CaO	1·7	7·0	7·2	6·2	8·6
Na_2O	5·5	3·4	3·2	4·2	4·0
K_2O	4·5	1·7	2·0	2·3	1·7
H_2O-	0·81	1·8	1·6	1·8	0·6
H_2O+	1·4	1·3	1·3	1·5	0·8
TiO_2	1·1	3·4	3·5	3·3	2·9
P_2O_5	0·25	0·59	0·51	0·61	0·9
MnO	0·09	0·20	0·17	0·19	0·2
CO_2	<0·05	<0·05	<0·05	<0·05	—
Total	100·0	99·7	99·7	99·8	99·9
		Molecular norms			
Q	5·2	3·2	3·3	—	
C	2·6	0·3	—	—	
Or	26·6	10·1	11·9	13·6	
Ab	46·5	28·9	27·2	35·6	
An	6·8	31·0	30·6	26·5	
Ne	—	—	—	—	
Mt	3·2	—	2·3	—	
Hm	2·5	6·5	3·7	8·3	
Il	2·1	6·4	6·7	2·7	
Ap	0·6	1·4	1·2	1·4	
Ol	—	—	—	—	
Di	—	—	1·5	—	
Hy	1·7	9·2	8·8	6·5	

* Present address: Stanford University, Stanford, California.

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activity that produced the numerous seamounts on the ridge[10].

The rocks in the vicinity of Chatham Bay, like those studied by Chubb at Wafer Bay, include flows, tuff and agglomerates. East and south of Chatham Creek the flows are thick, massive and nearly horizontal and seem to extend several kilometres along the coast. These flows are capped by a thick sequence of volcanic ash in the basin at the head of Chatham Creek. In contrast, the flows north of Chatham Creek are thin and are interbedded with crystal lithic tuff probably of subaerial origin. Most of these flows dip seaward away from present topographic highs, suggesting initial dip rather than deformation. The northern tip of Cocos consists of bedded tuffs that dip northward. Nuez Island is composed primarily of agglomerates, which also dip northward away from the central mass of Cocos, but there the dips are sufficiently steep to suggest the possibility of some deformation.

diopsidic augite. The plagioclase phenocrysts are complexly twinned and strongly zoned; most of them are labradorite (An_{55-65}), but a few are as calcic as An_{80}. The groundmass is composed of plagioclase of composition An_{40-45}, olivine, diopsidic augite, iron–titanium oxides and apatite. Almost all of the olivine is altered to iddingsite. A modal analysis of sample 4*C*401 gave: 56 per cent plagioclase, 18 per cent pyroxene, 13 per cent olivine, 11 per cent opaque minerals and 1 per cent apatite. The chemical analyses of these three rocks also do not differ significantly (Table 1). They all contain less than 50 per cent silica, a relatively large amount of alumina and total alkalis, and are oxidized, as can be seen by their high Fe_2O_3/FeO ratios. More than 1·75 per cent titania is present, which is typical for oceanic basalts in contrast to circumoceanic basalts, which usually contain less TiO_2 (ref. 11).

Sample 4*C*400 from one of the thick massive flows is

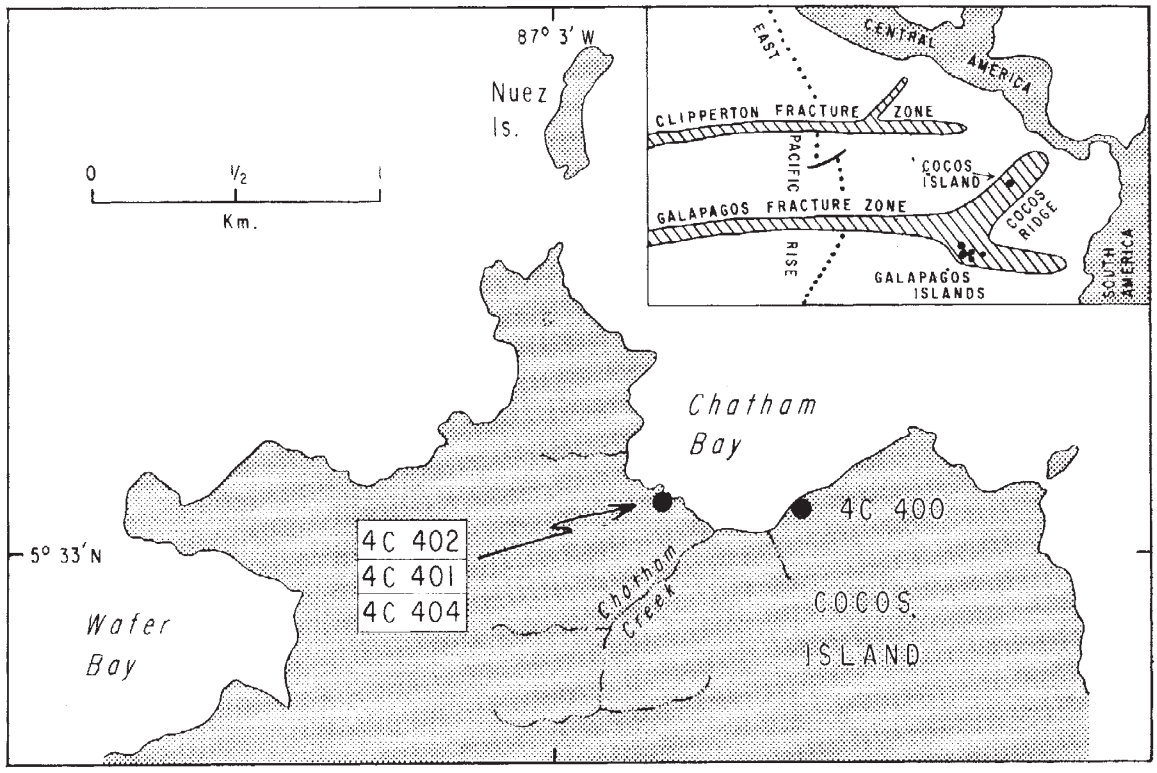


Fig. 1. Index map of northern Cocos Island showing locations of analysed samples. Samples north-west of Chatham Creek are shown in stratigraphic order. Inset shows the relationship of Cocos Island to the bathymetric features discussed in the text.

The marks of extremely active erosion characterize the present surface of Cocos Island. Despite the rapid growth of vegetation, many landslide scars are present in the vicinity of Chatham Bay. Mudflow and landslide deposits clog the lower reaches of the steeply incised canyon of Chatham Creek, and fans containing subangular boulders up to 2 m in diameter extend into the bay at the mouth of Chatham Creek, indicating that there has been recent mudflow activity.

For this study, samples were collected from thin flows north of Chatham Creek and from one of the thick, massive flows on the south side of Chatham Bay (Fig. 1). Samples 4*C*401, 4*C*402 and 4*C*404 from three of the thin flows are nearly identical alkali olivine basalts. These samples are porphyritic with a trachytic texture and contain phenocrysts of plagioclase, olivine and, rarely, a slightly porphyritic, granular trachyte that consists primarily of sodic plagioclase and sanidine with small amounts of olivine (altered to iddingsite), hypersthene, iron–titanium oxides and apatite. Petrographically and chemically, it resembles the specimen from Chatham Bay described by Merrill[3] as an andesite.

The alkali basalts which were studied are plotted on a K_2O–SiO_2 diagram (Fig. 2) together with previous analyses by Chubb[4]. The analyses all fall within the field of the alkali oceanic basalts[12]. No tholeiite has been reported from Cocos Island. The alkali basalts compare closely in composition with average "oceanic" alkali basalt[12], and A. E. J. Engel and C. G. Engel[13] have pointed out that rocks of this composition predominate on the islands and the upper parts of seamounts associated with the East Pacific Rise. Petrologically, Cocos Island is thus

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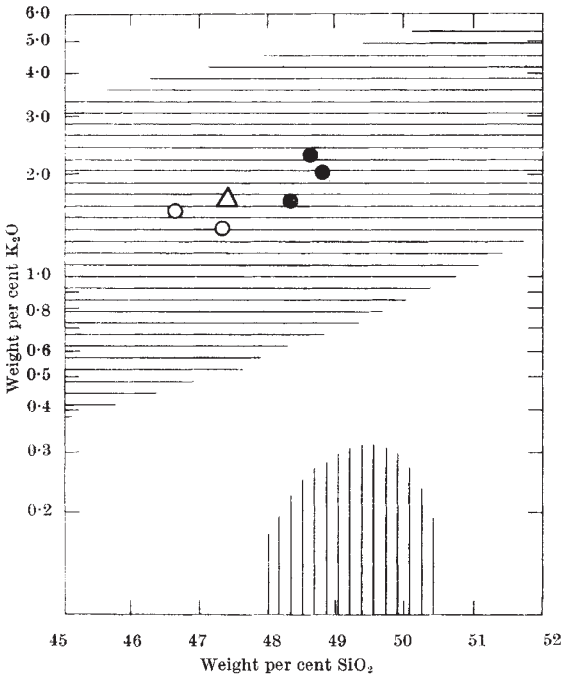


Fig. 2. K_2O–SiO_2 diagram for rocks from Cocos Island. Solid circles are analyses from this study, open circles are data from Chubb[4], and triangle is average oceanic alkali basalt[12]. Vertically and horizontally ruled areas are approximate fields of the oceanic tholeiite and alkali basalt[12].

oceanic, strengthening the conclusion that Cocos Ridge is not a submerged part of the continent.

The polarity of four or more oriented samples from each flow was measured in the field with a portable flux-gate magnetometer[14] and found to be internally consistent. These field measurements were confirmed by remeasuring the remanence of one sample from each flow in the laboratory[15] and checking its magnetic stability in alternating magnetic fields[16] (Table 2). The three flows north of Chatham Creek are reversely magnetized (Table 2). Two of these, flows 4*C*401 and 4*C*402, are underlain by baked tuffaceous material that is also reversely magnetized, indicating that the reversed remanent magnetism is a result of reversals of the Earth's field and not mineralogically controlled self reversal. The massive trachyte flow (4*C*400) is normally magnetized. The intensity of magnetization of the alkali olivine basalt samples is typical of basalt[17], whereas that of the trachyte (4*C*400) is much smaller. This is probably because the trachyte contains fewer grains of iron oxides.

An examination of thin sections showed that three of the four flows were suitable[18] for whole-rock potassium–argon age measurements. Argon analyses were by standard isotope dilution techniques using methods and techniques previously described[19]. Potassium was determined by flame photometry using a lithium internal standard. The ages (Table 3) of flows 4*C*401 and 4*C*402 are consistent with their stratigraphic relations, and the ages of all three flows are tightly grouped near 2×10^6 yr.

Recent studies of the history of geomagnetic reversals indicate that the polarity structure consists of polarity epochs, which are periods of time during which the Earth's magnetic field was predominantly or entirely of one polarity, and polarity events, which are like epochs but of much shorter

Table 2. PALAEOMAGNETIC DATA FOR ROCKS FROM COCOS ISLAND

Sample No.	Original magnetization D	I	J	After partial demagnetization D	I	J	Polarity
4*C*400	003°	21°	$8{\cdot}02 \times 10^{-4}$	008°	24°	$2{\cdot}97 \times 10^{-4}$	Normal
4*C*401	182°	00°	$2{\cdot}51 \times 10^{-3}$	182°	06°	$1{\cdot}61 \times 10^{-3}$	Reversed
4*C*402	173°	−07°	$2{\cdot}41 \times 10^{-3}$	173°	−08°	$1{\cdot}20 \times 10^{-3}$	Reversed
4*C*404	170°	−01°	$4{\cdot}27 \times 10^{-3}$	170°	−01°	$2{\cdot}30 \times 10^{-3}$	Reversed

D, Declination; I, inclination; J, intensity in emu/cc.

Partial demagnetization was in a.c. field of 400 gauss.

Table 3. ANALYTICAL DATA FOR POTASSIUM–ARGON AGE DETERMINATIONS ON VOLCANIC ROCKS FROM COCOS ISLAND, COSTA RICA

Sample No.	K_2O (per cent)	Sample weight (g)	Argon Ar^{40}_{rad}/g (moles)	$\frac{Ar^{40}_{rad} \times 100}{Ar^{40}_{total}}$	Calculated age* (10^6 yr)
4*C*400	4·78, 4·81	9·956	$1{\cdot}480 \times 10^{-11}$	62·8	$2{\cdot}09 \pm 0{\cdot}04$
4*C*401	2·12, 2·13	10·000	$0{\cdot}637 \times 10^{-11}$	25·9	$2{\cdot}03 \pm 0{\cdot}06$
4*C*402	2·22, 2·22	8·624	$0{\cdot}632 \times 10^{-11}$	19·0	$1{\cdot}93 \pm 0{\cdot}09$

* $\lambda_\varepsilon = 0{\cdot}585 \times 10^{-10}$ yr^{-1}, $\lambda_\beta = 4{\cdot}72 \times 10^{-10}$ yr^{-1}, $K^{40}/K = 1{\cdot}19 \times 10^{-4}$ mole/mole.

The ± values are estimates of the analytical precision at the 68 per cent confidence level[28].

Analyses were made on whole-rock samples.

duration[20–22]. One of the best documented events is the Olduvai normal event which occurred about 2·0 to $1{\cdot}9 \times 10^6$ yr ago. The normal polarity of this event is recorded (Fig. 3) in lava flows from Olduvai Gorge, Tanzania[23–24], the Pribilof Islands, Alaska[25], Réunion Island in the Indian Ocean[26] and Victoria, Australia[27]. From the polarity and age of sample 4*C*400, Cocos Island also appears to be a locality where volcanism occurred during the Olduvai event. The duration of the Olduvai event has been estimated to be about $0{\cdot}14 \times 10^6$ yr (refs. 28 and 29), whereas the spread in potassium–argon ages is nearly twice this. Although undoubtedly partly caused by dating errors, the spread in ages suggests that the Olduvai event consists of multiple, widely spaced shorter events, for which some evidence has already been found in deep-sea sediment cores[29].

In a similar study on the Galapagos Islands[30] we found that volcanism began at least $2{\cdot}4 \times 10^6$ yr ago and has continued to the present time. Thus there is at least some overlap between the volcanic activity on the Galapagos Fracture Zone and that on the Cocos Ridge. On

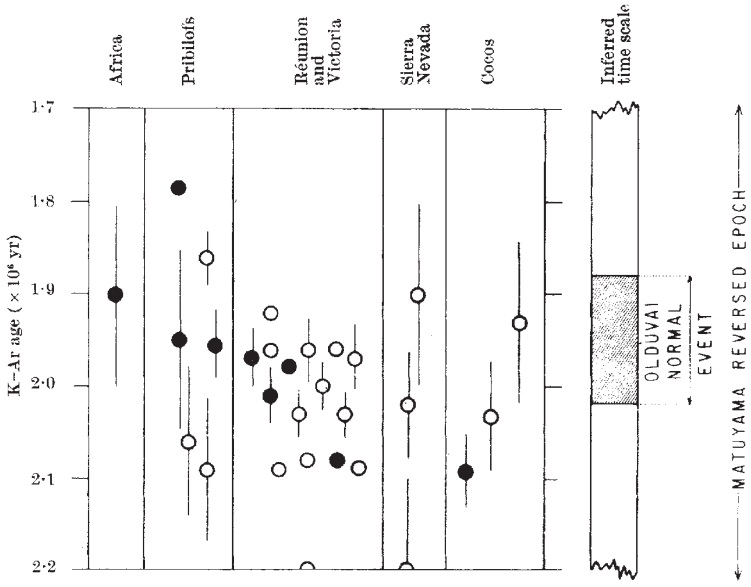


Fig. 3. Comparison of data from Cocos Island rocks with worldwide palaeomagnetic–radiometric data from rocks having ages between 1·7 and $2{\cdot}2 \times 10^6$ yr (refs. 23–27). Solid circles indicate normal polarity, open circles indicate reversed polarity, and vertical bars are estimates of standard deviation of analytical error where available.

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both Cocos and the Galapagos Islands, it is possible that older rocks may exist beneath a surface cover of younger flows, so that the time when individual islands first emerged from the sea cannot be determined with certainty. The geological evidence, however, does not contradict the idea that Cocos Island may have served as a stepping stone for organisms migrating to the Galapagos Islands, even though it is unlikely that the Cocos Ridge was ever a land bridge.

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Extragalactic Soft X-ray Astronomy

by
M. J. REES
D. W. SCIAMA
G. SETTI*
Department of Applied Mathematics and Theoretical Physics,
Silver Street,
Cambridge
and
Institute of Theoretical Astronomy,
University of Cambridge

The opacity of the galaxy to soft X-rays may be less than previously supposed. This may permit a variety of extragalactic objects to be observed at ~ 50 Å. Some implications of this situation are discussed here.

We wish to discuss in this article some astrophysical implications of soft X-ray observations. Our discussion is motivated by new evidence which indicates that the galaxy may be more transparent to X-rays than was previously believed, and that extragalactic radiation of wavelength ~ 50 Å (or even longer) may penetrate to us without severe attenuation.

The X-ray opacity of the galaxy is straightforwardly related to the column density of matter along the line of sight. It has traditionally been thought that 21 cm radio measurements yield a direct measure of the column density of hydrogen atoms: the optical depth of the galaxy to X-rays can then be calculated, making allowance for contributions from molecular hydrogen, from helium, and from heavy elements in interstellar space. This calculation leads to the conclusion that the optical depth at 50 Å $\gtrsim 1$ in almost all directions. There is now, however, some evidence which suggests that the amount of hydrogen in interstellar space may be significantly lower than the value inferred from 21 cm data. Some recently obtained ultraviolet spectra of stars in the direction of Orion reveal that the Lyman α absorption line is surprisingly narrow[1]. Fischel and Stecher[2] have made a proposal which may resolve the discrepancy. They show that "pumping" by blue-shifted Lyman α radiation can lead to population inversion in the hyperfine states of atomic hydrogen. Because 21 cm radiation would be amplified by the ensuing maser action, a lower column density of hydrogen is then needed to produce a given intensity in the 21 cm line.

Further evidence that the interstellar hydrogen density may be lower than the conventional estimates was recently reported by Bowyer, Field and Mack[3], who have measured the X-ray flux in the 44–70 Å range over a wide area of sky. They have discovered a background flux, stronger towards the Galactic Pole, which probably originates outside the galaxy. If this flux is truly isotropic (as is the X-ray background at shorter wavelengths), the observed latitude dependence indicates that the opacity of the galaxy should be about three times lower than the usual estimates. In consequence, the optical depth at 50 Å would be $\lesssim 1$ over a wide area of sky at high galactic latitudes. Although the optical depth goes roughly as λ^3, observations at even 100 Å may be feasible in certain directions where there are "windows" in the hydrogen

* On leave from the Institute of Physics, University of Bologna.

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